

# WTAB-R

## WEIGHT INDICATOR

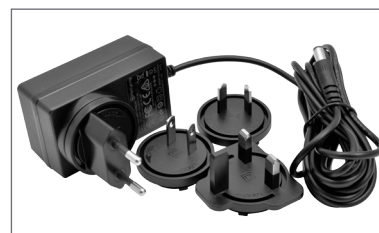
LAUMAS®



D-SUB connectors - IP40



Integrated thermal printer (on request)



Universal power supply included  
24 VDC/1 A - 100÷240 VAC input  
3 m cable length

### CERTIFICATIONS

-  OIML R76:2006, class III, 3x10000 divisions, 0.2  $\mu$ V/VSI / OIML R61 - WELMEC Guide 8.8:2011 (MID)
-  UL Recognized component - Complies with United States and Canada regulations
-  Complies with the Eurasian Customs Union regulations
-  Equivalent of the CE marking for the United Kingdom
-  NMI Trade Approved - Complies with Australian market regulations for legal for trade use
-  Complies with New Zealand regulations for legal for trade use
-  NTEP -  $n_{max}$  10000 - Class III/IIIL - Complies with United States regulations for legal for trade use
-  Complies with Chinese market regulations for legal for trade use

#### CERTIFICATIONS ON REQUEST

-  Conformity assessment (initial verification) in combination with Laumas weighing module
-  Complies with the regulations of the Russian Federation for legal for trade use

### FIELD BUSES

**MODBUS RTU**  
**MODBUS/TCP**

**CANopen**

**PROFIBUS**

**DeviceNet**

**EtherNet/IP**

**ETHERNET**  
**TCP/IP**

**PIV** CERTIFIED  
PROFIBUS • PROFINET

### DESCRIPTION

- ABS desk weight indicator.
- Dimensions: 315x170x315 mm.
- 6-digit semi-alphanumeric red LED display (20 mm height) - 16 signaling LED.
- 8-key keyboard.
- IP40 protection rating.
- Real-time clock/calendar with buffer battery.
- Power supply included.
- D-SUB connectors.
- The instrument can be configured and managed using the free "Instrument Manager" PC software, which you can download from [www.laumas.com](http://www.laumas.com).

### INPUTS/OUTPUTS AND COMMUNICATION

- RS485/RS232 serial ports for communication via protocols ModBus RTU, ASCII Laumas or continuous one way transmission.
- 5 relay outputs controlled by the setpoint values or via protocols (4 outputs if analog output is present).
- 3 optoisolated PNP digital inputs: status reading via serial communication protocols (2 inputs if analog output is present).
- 1 load cell dedicated input.
- Current or voltage 16 bit optoisolated analog output (option on request).
- WiFi module (option on request).

### MAIN FUNCTIONS

- Connections to:
  - PLC via analog output (on request);
  - PC/PLC via RS485/RS232 (up to 99 instruments with line repeaters, up to 32 without line repeaters);
  - remote display and printer via RS485/RS232;
  - up to 8 load cells in parallel by junction box;
  - intelligent junction box or other multichannel instruments: allow the use of advanced functions as digital equalization, load distribution analysis and automatic diagnostics.
  - IoT gateway for cloud connection via RS485.
- TCP/IP WEB APP: integrated software in combination with the WiFi module and Ethernet TCP/IP options for remote supervision, management and control of the instrument.
- Piece counting.
- Weight totalizing.
- Digital filter to reduce the effects of weight oscillation.
- Theoretical calibration (via keyboard) and real calibration (with sample weights and the possibility of weight linearization up to 8 points).
- Tare weight zero setting.
- Automatic zero setting at power-on.
- Gross weight zero tracking.
- Semi-automatic tare (net/gross weight).
- 9 preset tare values that can be stored.
- Semi-automatic zero.
- Displaying of the maximum weight value reached (peak).
- Direct connection between RS485 and RS232 without converter.
- Hysteresis and setpoint value setting.
- 12 groups selection by 5 setpoint via external selector switch or contact (option on request).
- Weight value printing with date and time via keyboard or external contact.
- Printing (header) can be customized using the free "JollyPrint" PC software, available at [www.laumas.com](http://www.laumas.com).
- The indicator can be used as a remote display with setpoints.
- Labeling machine management.

### Approved versions for legal for trade use

- System parameters management protected by qualified access via software (password), hardware or fieldbus.
- Weight subdivisions displaying (1/10 e).
- Three operation mode: single interval or multiple ranges or multi-interval.
- Net weight zero tracking.
- Calibration.
- Alibi memory (option on request).
- The following values can be printed via keyboard or external contact: gross weight, net weight, tare, preset tare, date, time, ID code (alibi memory).

### TECHNICAL FEATURES

|                                                                  |                                                                                                    |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Power supply and consumption                                     | 12÷24 VDC ±10%; 6 W                                                                                |
| Number of load cells • Load cells supply                         | up to 8 (350 Ω) - 4/6 wires • 5 VDC/120 mA                                                         |
| Linearity • Analog output linearity                              | <0.01% full scale • <0.01% full scale                                                              |
| Thermal drift • Analog output thermal drift                      | <0.0005% full scale/°C • <0.003% full scale/°C                                                     |
| A/D Converter                                                    | 24 bit (16000000 points) - 4.8 kHz                                                                 |
| Divisions (with measurement range ±10 mV and sensitivity 2 mV/V) | ±999999 • 0.01 μV/d                                                                                |
| Measurement range                                                | ±39 mV                                                                                             |
| Usable load cells sensitivity                                    | ±7 mV/V                                                                                            |
| Conversions per second                                           | 300                                                                                                |
| Display range                                                    | ±999999                                                                                            |
| Decimals • Display increments                                    | 0÷4 • x1 x2 x5 x10 x20 x50 x100                                                                    |
| Digital filter • Readings per second                             | 10 levels • 5÷300                                                                                  |
| Relay outputs                                                    | 5/4 - max 115 VAC/150 mA                                                                           |
| Optoisolated digital inputs                                      | 3/2 - 5÷24 VDC PNP                                                                                 |
| Serial ports                                                     | RS485, RS232                                                                                       |
| Baud rate                                                        | 1200, 2400, 4800, 9600, 19200, 38400, 115200 (bit/s)                                               |
| Optoisolated analog output (option on request)                   | 16 bit = 65535 divisions. 0÷20 mA; 4÷20 mA (up to 300 Ω)<br>0÷10 V; 0÷5 V; ±10 V; ±5 V (min 10 kΩ) |
| Humidity (condensate free)                                       | 85%                                                                                                |
| Storage temperature                                              | -30 °C +80 °C                                                                                      |
| Working temperature                                              | -20 °C +60 °C                                                                                      |

|                                                                                     |                                                                  |                                 |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------|
|  | Relay digital outputs                                            | 5/4 - max 30 VAC, 60 VDC/150 mA |
|                                                                                     | Working temperature                                              | -20 °C +50 °C                   |
|                                                                                     | Equipment to be powered by 12-24 VDC LPS or Class 2 power source |                                 |

| METROLOGICAL SPECIFICATIONS OF<br>TYPE-APPROVED INSTRUMENTS | OIML                                                                                                                                                                                                                                                                                                                                 | NTEP                                              |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Applied standards by region                                 | EU: 2014/31/UE; OIML R76:2006; EN45501:2015<br>Russian Federation: GOST OIML R76-1-2011<br>United Kingdom: Non-automatic Weighing Instrument Regulations 2016<br>Australia: National Measurement Regulations 1999<br>New Zealand: Weights and Measures Regulations 1999<br>China: Law on Metrology of the People's Republic of China | USA: NIST HANDBOOK 44, 2020;<br>NCWM PUB 14, 2021 |
| Operation modes                                             | single interval, multi-interval, multiple range                                                                                                                                                                                                                                                                                      | single interval, multi-interval, multiple range   |
| Accuracy class                                              | III or IIIL                                                                                                                                                                                                                                                                                                                          | III or IIIL                                       |
| Maximum number of scale verification divisions              | 10000 (class III); 1000 (class IIIL)                                                                                                                                                                                                                                                                                                 | 10000 (class III/IIIL)                            |
| Minimum input signal for scale verification division        | 0.2 μV/VSI                                                                                                                                                                                                                                                                                                                           |                                                   |
| Working temperature                                         | -10 °C +40 °C                                                                                                                                                                                                                                                                                                                        | -10 °C +40 °C (+14 °F +104 °F)                    |

### OPTIONS ON REQUEST

#### POWER SUPPLY



12.2 V rechargeable lead battery, 2.8 Ah capacity, supplied already installed in the instrument.  
Operating time: 16 hours.

#### ACCESSORIES



Integrated thermal printer: 24 column, paper end sensor, working temperature: 0÷50 °C, humidity: 20%÷80%, paper roll included (width: 57 ±0.5 mm - outside diameter: 50 mm).  
→ *RS485 port not available.*



Thermal paper roll.



Adhesive thermal paper roll.

#### INTERFACES AND FIELDBUSES



★ **WiFi module** (2.4 GHz) for wireless connection via integrated web server (for remote supervision, management and control of the instrument) or via ModBus RTU, ASCII Laumas protocols.



★ Optoisolated 16 bit **analog output**.  
→ *One input and one output not available.*



★ **Additional RS485 port**.  
→ *One input and one output not available.*



★ **CANopen** protocol.



★ **DeviceNet** protocol.



★ **Profibus DP** protocol.



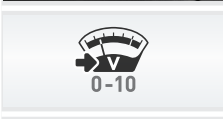
★ **Ethernet/IP** protocol - Ethernet port.



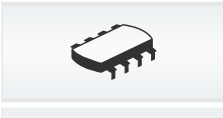
★ **Ethernet TCP/IP** protocol - Ethernet port.  
Integrated software for remote supervision, management and control of the instrument.

★ *Select one option among those marked with an asterisk.*

### OPTIONS ON REQUEST

|                                                                                   |                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | * <b>Modbus/TCP</b> protocol - Ethernet port.                                                                                                                                                                 |
|  | * <b>PROFINET IO</b> protocol - Ethernet port.                                                                                                                                                                |
|  | <b>DATAUSB</b><br><b>USB</b> port for data storage to pen drive (included).<br>These data (weighed values, alarms) can be imported and processed on the PC using the PROG-DB software included in the supply. |
|  | Weight reading from 0-10 VDC input (15 kΩ).                                                                                                                                                                   |
|  | Weight reading from 4-20 mA input (120 Ω).                                                                                                                                                                    |

### APPLICATIONS - SOFTWARE

|                                                                                     |                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Alibi memory.                                                                                                                                                                                                                                                                                                            |
|  | <b>DATAPC</b><br>Data transfer from the instrument to the PC, via RS232 (directly) or RS485 (by converter) serial port. These data (weighed values, alarms) can be imported and processed on the PC using the PROG-DB software included. We suggest to use this option when the indicator is always connected to the PC. |

\* Select one option among those marked with an asterisk.

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